

AI PROJECT

Smart Study Planner

AI-Based Personalized Study Scheduling Agent

Sarfraz Ali Katpar (023-23-0045)

Amjad Ali Abro (023-23-0043)

Supervisor: Dr Muhammad Ismail Mangrio

The Problem Statement



Poor Time Management

Students can't estimate how much time each subject needs



No Prioritization

All subjects treated equally regardless of difficulty or exam date



No Adaptability

Existing tools don't reschedule when study sessions are missed

Our Solution: An AI-powered system that automates scheduling and prioritization

System Overview



AI Planner

Subject input form + ML prediction +
Schedule generation + CSV export



Dashboard

Metric cards + Progress overview +
Streak tracker + Weekly chart



My Subjects

Progress tracking + Reschedule missed
sessions + Remove subjects

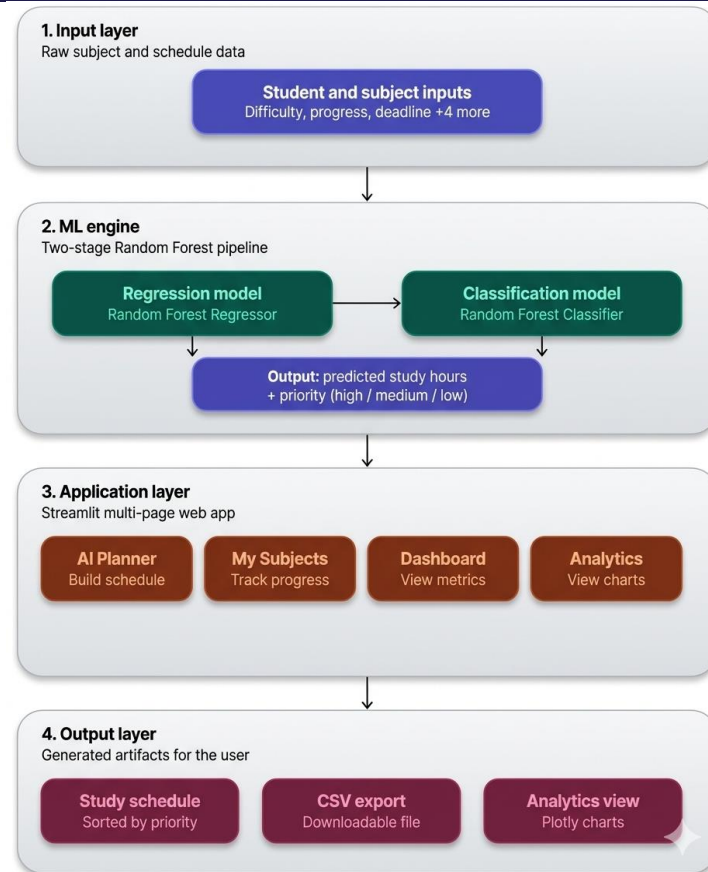


Analytics

Priority pie chart + Completion bar chart
+ Schedule summary table

← 4 Pages
→

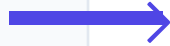
System Architecture



Machine Learning — The Brain of Agent

Input Features

- Subject Difficulty
- Total Chapters
- Chapters Done
- Days Until Exam
- Daily Free Hours
- Mid Term Score
- Urgency Score



Regression Model

Random Forest Regressor

Output: "This subject needs 8.5 study hours"

↓ *hours fed as extra feature* ↓

Classification Model

Random Forest Classifier

Output: ● HIGH / ● MEDIUM / ● LOW priority

Technology Stack



Python 3.10

Core Language



Streamlit

Web-Based
Framework



Scikit-learn

For ML Models



Pandas

For Data Processing



Plotly

For Interactive Charts



joblib

Model Serialization

Results & Conclusion

- Fully functional 4-module web application
- 2 trained ML models integrated into UI
- Real-time schedule generation with priority
- Interactive Plotly analytics charts
- CSV export for study schedules
- Adaptive rescheduling when sessions are missed

Future Enhancements

- Database for persistent storage
- User login & authentication
- Mobile-friendly layout
- Pomodoro timer integration
- Google Calendar sync

Thank You!

Project Demo